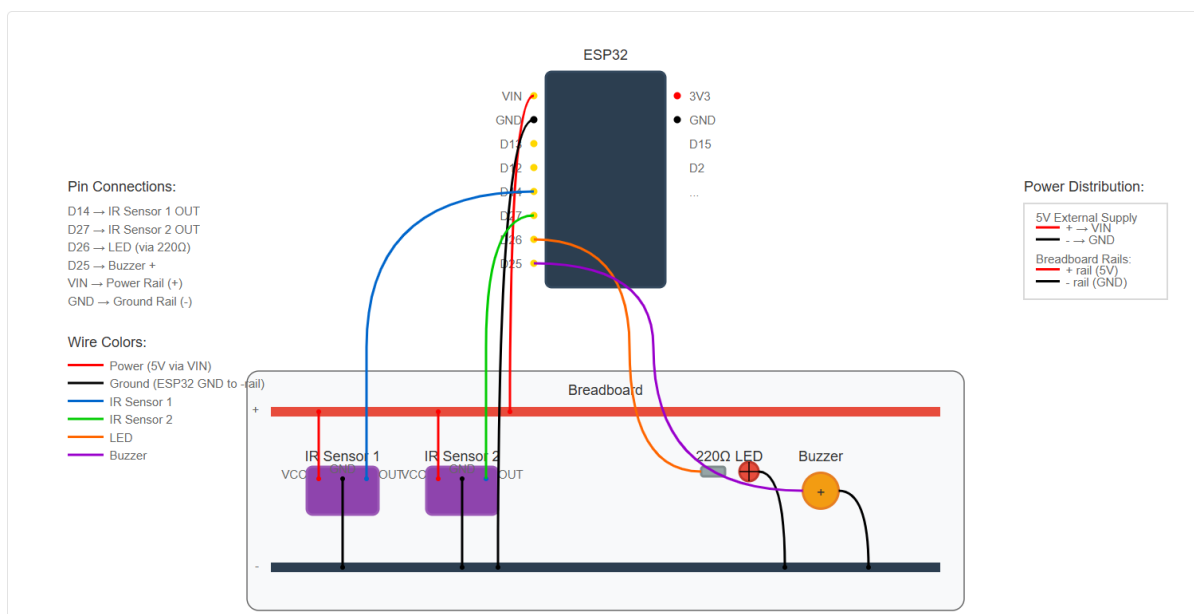


Bidirectional Visitor Counter using ESP32, IR Sensors, Buzzer, LED and Blynk App

Project Report and Documentation

1 Circuit Diagram

ESP32 Bidirectional Visitor Counter Circuit Diagram



1.1 Component List

- ESP32 Development Board
- 2× IR Obstacle Avoidance Sensors
- 1× LED (any color)
- 1× 220 Resistor
- 1× Active Buzzer (3-5V)
- 1× Breadboard
- Jumper wires (Male-to-Male and Male-to-Female)

1.2 Pin Connection Summary

Component	Pin	ESP32 Pin
IR Sensor 1	OUT	D14
IR Sensor 2	OUT	D27
LED	Anode (+)	D26 (via 220 resistor)
Buzzer	Positive (+)	D25
All Components	VCC/+	3.3V
All Components	GND/-	GND

Table 1: Pin Connection Table

2 Arduino Code

```

1 #define BLYNK_PRINT Serial
2
3 #define BLYNK_TEMPLATE_ID "TMPL6fTt5tuUg"
4 #define BLYNK_TEMPLATE_NAME "Bidirectional Visitor Counter"
5 #define BLYNK_AUTH_TOKEN "VXuRa-08gETyth0bwScmeDtLONASoAwZ"
6
7 #include <WiFi.h>
8 #include <BlynkSimpleEsp32.h>
9
10 // Blynk Auth Token
11 char auth[] = "VXuRa-08gETyth0bwScmeDtLONASoAwZ";
12 // WiFi credentials - Use your home's WiFi name and password here
13 char ssid[] = "kingFisher";
14 char pass[] = "BortY8728@";
15
16 // Pin Configuration
17 #define IR_SENSOR_1 14
18 #define IR_SENSOR_2 27
19 #define LED_PIN 26
20 #define BUZZER_PIN 25
21
22 // Variables
23 int count = 0;
24 const int threshold = 10; //for sound buzzer
25 bool lastIR1 = HIGH, lastIR2 = HIGH;
26 unsigned long lastTriggerTime = 0;
27 const unsigned long debounceDelay = 300; // ms
28
29 void setup() {
30   Serial.begin(115200);
31   pinMode(IR_SENSOR_1, INPUT_PULLUP);
32   pinMode(IR_SENSOR_2, INPUT_PULLUP);
33   pinMode(LED_PIN, OUTPUT);
34   pinMode(BUZZER_PIN, OUTPUT);
35   digitalWrite(LED_PIN, LOW);

```

```
36  digitalWrite(BUZZER_PIN, LOW);
37
38  Blynk.begin(auth, ssid, pass);
39 }
40
41 void loop() {
42   Blynk.run();
43
44   bool ir1 = digitalRead(IR_SENSOR_1) == HIGH;
45   bool ir2 = digitalRead(IR_SENSOR_2) == HIGH;
46
47   enum State { IDLE, IR1_TRIGGERED, IR2_TRIGGERED } ;
48   static State state = IDLE;
49   static unsigned long firstTriggerTime = 0;
50   const unsigned long sequenceTimeout = 10000;
51
52   unsigned long now = millis();
53
54   switch (state) {
55     case IDLE:
56       // Start entry sequence if IR1 triggers
57       if (ir1 && !ir2) {
58         state = IR1_TRIGGERED;
59         firstTriggerTime = now;
60       }
61       // Start exit sequence if IR2 triggers
62       else if (ir2 && !ir1) {
63         state = IR2_TRIGGERED;
64         firstTriggerTime = now;
65       }
66       break;
67
68     case IR1_TRIGGERED:
69       // Wait for IR2 to be triggered within the timeout for
        entry
70       if (ir2 && now - firstTriggerTime < sequenceTimeout) {
71         count++;
72         Serial.println("Entry detected");
73         updateDevices();
74         state = IDLE;
75       }
76       // If sensors released or timeout passed, reset
77       else if ((!ir1 && !ir2) || (now - firstTriggerTime >=
        sequenceTimeout)) {
78         state = IDLE;
79       }
80       break;
81
82     case IR2_TRIGGERED:
83       // Wait for IR1 to be triggered within the timeout for exit
84       if (ir1 && now - firstTriggerTime < sequenceTimeout) {
```

```
85     if (count > 0) count--;
86     Serial.println("Exit detected");
87     updateDevices();
88     state = IDLE;
89 }
90 // If sensors released or timeout passed, reset
91 else if ((!ir1 && !ir2) || (now - firstTriggerTime >=
92     sequenceTimeout)) {
93     state = IDLE;
94 }
95 break;
96 }
97
98
99 void updateDevices() {
100     // Light control
101     digitalWrite(LED_PIN, count > 0 ? HIGH : LOW);
102     // Buzzer control
103     if (count >= threshold) {
104         digitalWrite(BUZZER_PIN, HIGH);
105         delay(500);
106         digitalWrite(BUZZER_PIN, LOW);
107     }
108     // Blynk updates
109     Blynk.virtualWrite(V0, count); // Value Display widget
110     Blynk.virtualWrite(V1, count > 0 ? 255 : 0); // LED widget
111 }
112
113 BLYNK_WRITE(V2) {
114     if (param.asInt() == 1) {
115         count = 0;
116         updateDevices();
117         Serial.println("Count reset via Blynk image button");
118     }
119 }
```

3 Code Explanation

- **Blynk and WiFi Setup:** Auth token, WiFi SSID, and password are initialized.
- **Pin Definitions:** IR sensors on D14 and D27, LED on D26, and buzzer on D25.
- **Visitor Logic:** Uses a finite state machine (FSM) to detect entry and exit sequences based on IR sensor trigger order.
- **updateDevices():** Updates LED, buzzer, and sends virtual pin values to Blynk app.
- **BLYNK_WRITE():** Resets count if virtual pin V2 is pressed from Blynk image button.

4 Hardware Connections (ESP32 + Breadboard)

ESP32 Pin Layout

- **EN side (connected to breadboard):** VIN, GND, D13, D12, D14, D27, D26, D25
- **BOOT side (open):** 3V3, GND, D15, D2, etc.

Wiring Instructions

1. Power

- 3.3V (BOOT side) → Breadboard positive rail (male-to-female wire)
- GND (BOOT side) → Breadboard negative rail (male-to-female wire)

2. IR Sensor 1 (connected to D14)

- VCC → Breadboard + rail
- GND → Breadboard - rail
- OUT → D14 (male-to-male)

3. IR Sensor 2 (connected to D27)

- VCC → Breadboard + rail
- GND → Breadboard - rail
- OUT → D27 (male-to-male)

4. LED (connected to D26)

- Anode (+, longer leg) → Resistor → D26 (male-to-male)
- Cathode (-, shorter leg) → GND rail

5. Buzzer (connected to D25)

- + → D25 (male-to-male)
- - → GND rail

Jumper Wire Types

- Male-to-Male: Breadboard to ESP32 inserted pins
- Male-to-Female: ESP32 BOOT side pins to breadboard power rails

5 Project Features

5.1 Bidirectional Detection

The system uses two IR sensors positioned in sequence to detect the direction of movement:

- **Entry:** IR Sensor 1 triggered first, then IR Sensor 2
- **Exit:** IR Sensor 2 triggered first, then IR Sensor 1

5.2 Visual and Audio Feedback

- **LED Indicator:** Turns ON when count $\neq 0$, OFF when count = 0
- **Buzzer Alert:** Activates when visitor count reaches threshold (10 people)

5.3 IoT Integration

- **Blynk App:** Real-time monitoring via smartphone
- **Virtual Pins:** V0 (count display), V1 (LED status), V2 (reset button)
- **Remote Reset:** Count can be reset remotely via Blynk app

6 Blynk App Configuration

6.1 Virtual Pin Setup

Virtual Pin	Widget	Function
V0	Value Display	Shows current visitor count
V1	LED Widget	Visual indicator (ON/OFF)
V2	Button Widget	Reset counter to zero

Table 2: Blynk Virtual Pin Configuration

6.2 Required Blynk Credentials

- Template ID: TMPL6fTt5tuUg
- Template Name: Bidirectional Visitor Counter
- Auth Token: VXuRa-O8gETyth0bwScmeDtL0NASoAwZ

7 How It Works

7.1 Finite State Machine Logic

The system operates using a three-state finite state machine:

1. **IDLE State:** Waiting for sensor trigger
2. **IR1_TRIGGERED:** First sensor activated, waiting for second
3. **IR2_TRIGGERED:** Second sensor activated, waiting for first

7.2 Detection Sequence

- When a person approaches from outside:
 1. IR Sensor 1 detects person → State changes to IR1_TRIGGERED
 2. IR Sensor 2 detects person → Count increments → State returns to IDLE
- When a person exits from inside:
 1. IR Sensor 2 detects person → State changes to IR2_TRIGGERED
 2. IR Sensor 1 detects person → Count decrements → State returns to IDLE

7.3 Timeout Protection

A 10-second timeout prevents false counts from incomplete sequences or sensor noise.

8 Troubleshooting

8.1 Common Issues

- **Sensors not responding:** Check 3.3V power supply and ground connections
- **False counts:** Adjust sensor sensitivity potentiometers or increase timeout
- **Blynk connection issues:** Verify WiFi credentials and auth token
- **LED/Buzzer not working:** Check pin connections and component polarity

8.2 Calibration Tips

- Position sensors 50-80cm apart for optimal detection
- Adjust sensor height to waist level (80-100cm from ground)
- Test detection range by adjusting onboard potentiometers
- Ensure sensors face each other across the doorway

9 Future Enhancements

- Data logging with timestamps
- Email/SMS notifications when threshold is reached
- Web dashboard for historical data analysis
- Battery backup for power outages
- Multiple room monitoring with additional sensor pairs

The system provides an affordable and scalable solution for visitor monitoring in various applications including retail stores, offices, exhibitions, and smart home automation.